

A Closed Rank-One Formula for the Majority Adversary

Abstract

Let $k = 2m + 1$ be odd. We give a direct closed formula for the period-four rank-one, intersection-independent adversary for Majority. No separator matrix, eigenspace, Cauchy solve, or balancing parameter appears in the final answer. Define two explicit period-four degree weights p_t, q_t and take one finite Krawtchouk transform $W_k(a)$. Then

$$X_S[A, B] = \frac{2^{-n}}{2(k-1)!} W_k(|A \cap S|) W_k(|B \cap S|)$$

is feasible. The same equal normalization on the two Majority sides already has

$$\max_A \sum_S X_S[A, A] = \frac{4}{\pi} \log k + O(1).$$

Thus the sharp leading constant of the period-four rank-one witness survives in a particularly simple one-line formula.

1 The formula

Let

$$K_t(a) = \sum_j (-1)^j \binom{a}{j} \binom{k-a}{t-j}$$

be the binary Krawtchouk polynomial. Split the interior degrees into

$$Z_{01} = \{1 \leq r \leq k-1 : r \equiv 0, 1 \pmod{4}\}, \quad Z_{23} = \{1 \leq r \leq k-1 : r \equiv 2, 3 \pmod{4}\}.$$

Both sets have size m . For $0 \leq t \leq k$ define the two explicit degree weights

$$p_t = \prod_{r \in Z_{01}} (t-r), \quad q_t = \prod_{r \in Z_{23}} (t-r). \quad (1)$$

Their product is

$$p_t q_t = \prod_{r=1}^{k-1} (t-r). \quad (2)$$

In particular,

$$p_t q_t = 0 \quad (1 \leq t \leq k-1), \quad p_0 q_0 = (k-1)!.$$

Define one scalar count function

$$\boxed{W_k(a) = \sum_{t=0}^k K_t(a) (p_t + (-1)^t q_t), \quad 0 \leq a \leq k.} \quad (3)$$

Then the promised adversary matrices are simply

$$X_S[A, B] = \frac{2^{-n}}{2(k-1)!} W_k(|A \cap S|) W_k(|B \cap S|). \quad (4)$$

Thus every X_S is visibly positive semidefinite and has rank at most one.

The degree weights themselves can be written without products. With $(x)_N = x(x+1) \cdots (x+N-1)$, if $k = 4N + 1$ then

$$p_t = 16^N \left(\frac{1-t}{4} \right)_N \left(1 - \frac{t}{4} \right)_N, \quad (5)$$

$$q_t = 16^N \left(\frac{2-t}{4} \right)_N \left(\frac{3-t}{4} \right)_N. \quad (6)$$

If $k = 4N + 3$, then

$$p_t = -4^{2N+1} \left(\frac{1-t}{4} \right)_{N+1} \left(1 - \frac{t}{4} \right)_N, \quad (7)$$

$$q_t = -4^{2N+1} \left(\frac{2-t}{4} \right)_{N+1} \left(\frac{3-t}{4} \right)_N. \quad (8)$$

Equations (3) and (4), together with either (1) or (5)–(8), are a completely closed finite formula.

2 Why the formula automatically respects the Majority cut

Write

$$L(a) = \sum_{t=0}^k K_t(a) p_t, \quad U(a) = \sum_{t=0}^k K_t(k-a) q_t.$$

Since $K_t(k-a) = (-1)^t K_t(a)$,

$$W_k(a) = L(a) + U(a).$$

The useful point is that these two terms never overlap.

Lemma 2.1 (Degree becomes count support). *One has*

$$L(a) = 0 \quad (a > m), \quad U(a) = 0 \quad (a \leq m).$$

Hence

$$W_k(a) = \begin{cases} L(a), & 0 \leq a \leq m, \\ U(a), & m+1 \leq a \leq k. \end{cases} \quad (9)$$

Proof. The sequence $t \mapsto p_t$ is the restriction of a polynomial of degree m . Krawtchouk duality gives

$$L(a) = \frac{1}{\binom{k}{a}} \sum_{t=0}^k \binom{k}{t} K_a(t) p_t.$$

The Krawtchouk polynomial $K_a(t)$ is orthogonal, for the binomial weight, to every polynomial of degree $< a$. Thus $L(a) = 0$ when $a > m$. The same argument applied to q_t gives $\sum_t K_t(c) q_t = 0$ for $c > m$; taking $c = k - a$ proves the claim for U . $\square$

Therefore the single vector W_k already splits itself across the two Majority outputs: L lives on counts $0, \dots, m$, while U lives on counts $m+1, \dots, k$.

3 Feasibility in one calculation

Let concepts A, B have Johnson distance

$$d = k - |A \cap B| \geq 1.$$

For a uniformly random query S , put

$$a = |A \cap S|, \quad b = |B \cap S|.$$

The standard Krawtchouk correlation identity is

$$\mathbb{E}[K_s(a)K_t(b)] = \mathbf{1}\{s = t\} \binom{k-d}{t}. \quad (10)$$

On separating queries, one count is at most m and the other is at least $m + 1$. By Lemma 2.1,

$$W_k(a)W_k(b)\mathbf{1}\{f_A(S) \neq f_B(S)\} = L(a)U(b) + U(a)L(b).$$

Using $K_t(k-b) = (-1)^t K_t(b)$ and (10),

$$\mathbb{E}[L(a)U(b)] = \sum_{t=0}^k (-1)^t \binom{k-d}{t} p_t q_t.$$

By (2), every interior term $1 \leq t \leq k-1$ vanishes. The $t = k$ term vanishes because $d \geq 1$, while the $t = 0$ term equals $(k-1)!$. Thus

$$\mathbb{E}[L(a)U(b)] = (k-1)!.$$

The other orientation is identical, and hence

$$\mathbb{E} \left[\frac{W_k(a)W_k(b)}{2(k-1)!} \mathbf{1}\{f_A(S) \neq f_B(S)\} \right] = 1.$$

Multiplying by 2^{-n} and summing over all queries is exactly the dual adversary feasibility constraint. This proves:

Theorem 3.1 (Closed rank-one Majority witness). *For every odd k , the matrices (4) form a feasible, intersection-independent, rank-one solution of the Majority dual-adversary SDP.*

4 Cost and the logarithmic bound

Let

$$N_p = \sum_{t=0}^k \binom{k}{t} p_t^2, \quad N_q = \sum_{t=0}^k \binom{k}{t} q_t^2.$$

Krawtchouk orthogonality and the disjoint supports in (9) give the exact objective

$$\boxed{\mathcal{C}_k = \max_A \sum_S X_S[A, A] = \frac{N_p + N_q}{2(k-1)!}.} \quad (11)$$

So even the normalization is closed: no balancing scalar has to be solved for.

For completeness we record the asymptotic size. Put

$$E_p = |p_0 p_k|, \quad E_q = |q_0 q_k|, \quad S_p = \frac{N_p}{E_p}, \quad S_q = \frac{N_q}{E_q}.$$

Since $E_p E_q = ((k-1)!)^2$, if $r_k = E_p/(k-1)!$ then

$$C_k = \frac{1}{2} (r_k S_p + r_k^{-1} S_q). \quad (12)$$

The period-four products admit a standard Gamma-ratio evaluation. Define

$$\kappa = \frac{\Gamma(3/4)}{\Gamma(1/4)\sqrt{\pi}}, \quad \rho = \frac{\Gamma(1/4)}{\Gamma(3/4)\sqrt{\pi}}, \quad \kappa\rho = \frac{1}{\pi}.$$

Directly from (5)–(8), followed by the elementary harmonic convolution

$$\sum_{j=0}^N \frac{N+1}{(j+\alpha)(N-j+\beta)} = 2 \log N + O_{\alpha,\beta}(1),$$

one obtains

$$k = 4N + 1 : \quad S_q = 4\kappa \log k + O(1), \quad S_p = 4\kappa\rho^2 \log k + O(1), \quad (13)$$

$$k = 4N + 3 : \quad S_p = S_q = \frac{4}{\pi} \log k + O(1). \quad (14)$$

For $k = 4N + 3$, reflection exchanges the two period-four sets and $r_k = 1$. For $k = 4N + 1$,

$$r_k = \prod_{j=0}^{N-1} \frac{(4j+1)(4j+4)}{(4j+2)(4j+3)} = \rho^{-1} + O(N^{-1}). \quad (15)$$

Substituting (13)–(15) into (12),

$$C_k = \frac{1}{2} [\rho^{-1}(4\kappa\rho^2) + \rho(4\kappa)] \log k + O(1) = \frac{4}{\pi} \log k + O(1).$$

Together with (14) this proves

$$\boxed{C_k = \frac{4}{\pi} \log k + O(1).} \quad (16)$$

Remark 4.1 (How much simpler can $W_k(a)$ become?). *Formula (3) is a single finite Krawtchouk transform of completely explicit products. I do not see a further uniform reduction to one binomial or Gamma product in the count variable a . The transform contains real cancellations. For example, after removing a common integer factor, $W_k(a)$ begins as*

k	$(W_k(0), \dots, W_k(k))$
5	$(-2, 0, 1, 1, 0, 2),$
7	$(11, -1, 1, -3, -3, -1, -1, -11),$
9	$(-22, 0, -1, 0, 3, 3, 0, 3, 0, 18).$

Thus the one-sum expression (3) appears to be the natural simple closed form for the period-four witness. Importantly, the adversary matrix itself remains the one-line outer product (4).

Final closed form. For odd Majority, the whole witness is therefore

$$\boxed{X_S[A, B] = \frac{2^{-n}}{2(k-1)!} W_k(|A \cap S|) W_k(|B \cap S|), \quad W_k(a) = \sum_{t=0}^k K_t(a) (p_t + (-1)^t q_t).}$$

It is rank one by inspection; feasibility follows from $p_t q_t = 0$ on every interior degree; and its cost is $\frac{4}{\pi} \log k + O(1)$.